

Probabilistic electricity demand forecasting for buildings using a variational mode decomposition-attentive quantile network

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ABSTRACT

Accurate and reliable forecasting of building electricity demand is essential for energy management, demand response, and operational planning in smart buildings. However, building load profiles are influenced by interacting factors such as weather conditions, occupant behavior, and operational schedules, leading to nonstationary and multi-scale temporal patterns with inherent uncertainty that traditional deterministic models struggle to represent. To address these challenges, this study proposes a Variational Mode Decomposition (VMD)-Attentive Quantile Network for probabilistic building load forecasting, designed to jointly capture complex temporal dynamics and predictive uncertainty. The framework leverages VMD to separate load signals into frequency-specific components, enhancing robustness against nonstationary effects. A sequential GRU-LSTM encoder models complementary short-term fluctuations and long-term dependencies, while Multi-Head Attention dynamically weights influential time steps. An IMF-dependent gating mechanism suppresses high-entropy noise components. Quantile regression is incorporated to directly estimate conditional demand distributions and generate prediction intervals at multiple confidence levels. The model is evaluated on hourly electricity consumption data (2019–2023) from three functionally distinct campus buildings at National Taiwan University. Results indicate consistent outperformance of VMD-based benchmark models in point forecasting, with R^2 improvements of 8.4%–12.9% and reductions in MAE and RMSE of 16.9%–40.8% and 13.5%–39.6%, respectively. In probabilistic forecasting, the model produces more compact and reliable prediction intervals, reflected by lower PINAW and AIS values across confidence levels. Overall, the proposed approach offers a robust and interpretable solution for short-term building load forecasting across heterogeneous load patterns.

1. Introduction

This section introduces the research background, reviews relevant studies on building electricity demand forecasting, identifies key research gaps, and summarizes the main contributions of this study.

1.1. Background

Global energy demand continues to rise, and resource scarcity has intensified pressure on energy systems. As a major source of global energy use and carbon emissions, the building and construction sector consumes 32% of global energy and contributes 34% of global CO₂ emissions [1]. The sector has therefore become a critical focus for decarbonization and energy-efficiency improvement. Achieving this transition requires not only the deployment of low-carbon technologies,

but also continuous improvements in building operation and energy management [2].

In this context, accurate forecasting of building electricity demand plays an essential role in improving operational efficiency and supporting energy management in smart buildings [3,4]. Reliable short-term forecasts can support peak-demand mitigation, scheduling decisions, and more adaptive control of electricity use under time-varying operating conditions [5]. However, building electricity demand is strongly influenced by occupancy-related variability, operational schedules, and heterogeneous usage patterns. In commercial buildings, occupant-related interactions with building systems have been reported to account for approximately 27%–43% of total building energy consumption, making day-to-day demand behavior highly dynamic and difficult to predict [6]. These characteristics highlight the need for forecasting approaches that can better address the complexity of building electricity consumption in real operating environments [7].

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Nomenclature

LSTM	Long Short-Term Memory
Bi-LSTM	Bidirectional Long Short-Term Memory
GRU	Gated recurrent unit
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
VMD	Variational mode decomposition
IMF	Intrinsic Mode Function
MHA	Multi-Head Attention
TA	Temporal Attention
PINC	Prediction Interval Nominal Confidence
R^2	the coefficient of determination
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
PINAW	Prediction Interval Normalized Average Width
AIS	Average Interval Score
CERB	Civil Engineering Research Building
CEB	Civil Engineering Department Building
MDH	Ming-Da Hall

To address these challenges, data-driven forecasting methods have been widely adopted for building energy prediction. Existing studies have shown that machine learning and deep learning models can improve point forecasting accuracy by learning nonlinear and time-dependent patterns from operational data [8]. However, point forecasts alone may not fully support decision-making when future demand is uncertain. Probabilistic forecasting methods provide prediction intervals or predictive distributions, while decomposition-based strategies can help represent noisy and multi-scale load behavior [9,10]. These developments motivate the need for a forecasting framework that can jointly consider building load uncertainty and multi-scale temporal characteristics.

1.2. Literature review

This section reviews three streams of literature related to building electricity demand forecasting: deterministic models, probabilistic approaches, and decomposition-based methods. These studies provide the basis for positioning the proposed uncertainty-aware and multi-scale forecasting framework.

1.2.1. Deterministic building electricity demand forecasting

Data-driven models have been widely used for deterministic building energy and electricity demand forecasting because of their ability to capture nonlinear relationships between historical demand, temporal variables, weather conditions, and operational patterns. Runge and Zmeureanu [11] reviewed studies using artificial neural networks for forecasting building energy use and demand, showing that neural-network-based approaches have been extensively applied to building energy forecasting across different applications, input variables, forecasting horizons, and performance metrics. Chae et al. [12] introduced an ANN model with Bayesian regularization for sub-hourly electricity usage forecasting in commercial buildings, demonstrating the effectiveness of neural networks in capturing short-term building electricity patterns. Pham et al. [13] developed a machine-learning-based approach for predicting hourly energy consumption across multiple buildings, showing that deterministic models can support building energy efficiency and sustainability analysis. More recently, Somu et al. [14] proposed a CNN-LSTM framework for building energy consumption forecasting, combining convolutional feature extraction with temporal sequence learning to improve prediction accuracy. Li et al. [15] further developed a Bayesian optimized spatial-temporal attention

enhanced LSTM model for building energy prediction, indicating that attention-based recurrent architectures can improve the representation of seasonal, temporal, and peak energy patterns.

These studies demonstrate that deterministic data-driven models can effectively improve point forecasting accuracy in building energy and electricity demand prediction. However, they generally produce single-value forecasts and therefore provide limited information about the possible range of future demand. For building operation and energy management, this limitation is important because demand uncertainty, abrupt fluctuations, and irregular usage patterns cannot be fully represented by point predictions alone. This motivates the need for probabilistic forecasting methods that can quantify prediction uncertainty in addition to improving point accuracy.

1.2.2. Probabilistic building electricity demand forecasting

Load forecasting approaches can generally be divided into deterministic and probabilistic categories. Deterministic models generate single-value forecasts, whereas probabilistic forecasting provides prediction intervals or predictive distributions that describe possible future demand ranges [16]. This distinction is particularly important for building electricity demand forecasting because building loads are affected by weather conditions, occupancy patterns, operating schedules, and occasional peak events [17,18]. Compared with deterministic forecasts, probabilistic outputs can provide additional information for building energy management by describing not only the expected demand but also the uncertainty associated with future load variations [19].

Recent studies have increasingly applied probabilistic forecasting methods to building-level energy and electricity demand prediction. Segarra et al. [20] proposed a probabilistic load forecasting method for building energy models by accounting for uncertainty in forecast weather data, showing that probabilistic outputs can provide useful information for building-level energy modeling. Roth et al. [21] applied Bayesian probabilistic forecasting to residential smart meter prediction, demonstrating the value of uncertainty estimation in building-scale energy analysis. Seefoo Jarquin et al. [22] developed a short-term probabilistic load forecasting approach for university buildings using artificial neural networks, further showing the relevance of probabilistic forecasting in institutional building contexts.

Recent data-driven approaches have also explored interval and quantile-based forecasting for building load prediction. Liu et al. [23] developed a short-term probabilistic building load forecasting method based on feature selection, convolutional gated recurrent units, and quantile regression to represent building load uncertainty. Duan et al. [24] proposed an interval energy-forecasting method using LSTM and penalized quantile regression for building energy consumption prediction, showing that interval-based forecasts can provide more informative outputs than single-value estimates. These studies demonstrate that probabilistic forecasting provides a more comprehensive characterization of demand uncertainty than deterministic point prediction, making it valuable for building energy management and operational planning. Nevertheless, building electricity demand often remains difficult to model because of noisy, nonstationary, and multi-scale temporal behavior.

1.2.3. Decomposition-based methods for building load forecasting

To address noisy and multi-scale load behavior, decomposition-based methods have been widely adopted to improve signal regularity and separate frequency-dependent patterns before forecasting. Building demand often contains nonstationary fluctuations, abrupt disturbances, and frequency-dependent patterns that a single model may struggle to capture. To enhance signal regularity and mitigate noise, researchers frequently adopt decomposition methods such as Empirical Mode Decomposition (EMD) [25], Variational Mode Decomposition (VMD) [26], and Wavelet Transform (WT) [27].

Several building-focused studies have applied these methods as

preprocessing steps before forecasting. Zhang et al. [28] compared multiple DWT- and EMD-based time–frequency analysis techniques for noise cancellation in building energy data, using a real-building load forecasting case study. Their results show that noise-cancelled building energy data can improve data-driven load forecasting performance. Chou et al. [29] combined EEMD, Wavelet Transform, and LSTM to forecast building energy consumption across 20 buildings with different functions and locations, showing that decomposition can be used to represent building energy patterns before sequence learning. Zhang et al. [30] proposed a short-term power load forecasting method for industrial buildings using VMD for noise reduction, CEEMDAN for decomposition-reconstruction, and a TCN-Informer-BiGRU model for prediction. Chen et al. [31] developed a short-term building electricity load forecasting method using CEEMDAN decomposition, FFT-based frequency-domain analysis, K-means modal component grouping, and MKDCN-LSTM prediction, indicating that frequency-based grouping of decomposed components can improve the modeling of building load patterns.

These studies suggest that decomposition can improve the representation of noisy and nonstationary building load signals. However, most decomposition-based building forecasting methods still use decomposition primarily as a preprocessing step, after which the decomposed components are passed to downstream models. Although this strategy can simplify complex load patterns, the resulting components may exhibit different levels of regularity, noise, and predictability. Treating all components in the same manner may allow highly irregular components to interfere with the learned representation and reduce forecasting stability. Therefore, beyond decomposing the original building load signal, component-level refinement remains important for multi-scale building electricity demand forecasting.

Taken together, prior studies provide important foundations for building electricity demand forecasting through deterministic prediction, probabilistic uncertainty representation, and decomposition-based signal processing. However, these approaches are typically developed in isolation, and few studies integrate multi-scale decomposition, uncertainty-aware learning, and component-level refinement within a unified framework. In particular, existing decomposition-based methods often treat all decomposed components uniformly, without accounting for their heterogeneous levels of noise and predictability. This limitation can reduce model robustness when irregular components dominate the signal. These limitations motivate the development of an integrated framework that jointly enables multi-scale representation, uncertainty quantification, and adaptive component-level modeling for building electricity demand forecasting.

1.3. Research contributions

Building on these limitations, this study develops a VMD-Attentive Quantile Network for building electricity demand forecasting to jointly capture multi-scale temporal dynamics and predictive uncertainty within a unified framework. The model integrates VMD for signal decomposition, a hybrid GRU–LSTM encoder for temporal learning, and attention and gating mechanisms for adaptive feature refinement. A quantile regression output layer is incorporated to generate prediction intervals across multiple confidence levels, enabling forecasts that reflect both accuracy and uncertainty.

The main contributions of this study are summarized as follows:

- (1) A quantile-based probabilistic forecasting formulation is developed for building electricity demand, enabling the simultaneous generation of point forecasts and prediction intervals. By directly estimating multiple conditional quantiles, the proposed approach provides an explicit representation of demand uncertainty and supports interval evaluation across different confidence levels.
- (2) A multi-scale temporal learning architecture is proposed by integrating VMD-based decomposition, sequential GRU–LSTM

encoding, multi-head temporal attention, and entropy-guided IMF-dependent gating. This design enables the model to capture both short- and long-term dependencies while adaptively refining decomposed components based on their irregularity and predictability.

- (3) An integrated evaluation perspective for probabilistic building load forecasting is established, which jointly assesses point accuracy and interval quality under multiple confidence levels. By combining conventional error metrics with interval-based measures, the proposed approach enables a more comprehensive characterization of forecasting performance, encompassing both precision and uncertainty.

The remainder of this paper is structured as follows. Section 2 describes the datasets, preprocessing procedures, and methodological framework. Section 3 presents forecasting results, including point-based comparisons, interval analysis, and ablation studies. Section 4 concludes with key insights and future research directions.

2. Methodology and Materials

2.1. Research framework

The methodological framework of the proposed VMD-Attentive Quantile Network is illustrated in Fig. 1. It is designed to jointly improve point-forecasting accuracy and probabilistic-forecasting reliability for building electricity demand. The workflow consists of three major stages:

Step 1: Data Preprocessing.

Hourly electricity consumption data from three National Taiwan University campus buildings (2019–2023) serve as the modeling input. Missing values are imputed, and the series are normalized to ensure consistency across buildings. The preprocessed signal is then decomposed using VMD, yielding multiple IMFs. Each IMF captures patterns at different frequency bands, enabling multi-scale learning during downstream model training.

Step 2: Model Learning.

Each IMF acts as a channel in the model input and is sequentially processed by GRU and LSTM layers, which extract short-term fluctuations and long-term temporal dependencies, respectively. To emphasize informative temporal segments, a Multi-Head Attention mechanism assigns adaptive weights to different time steps. The resulting attention-aware features are projected through a shared fully connected layer to construct global latent representations.

These representations are distributed to K IMF-specific learning branches. Within each branch, a sample entropy mask evaluates IMF volatility, followed by a gating mechanism that suppresses noisy dimensions and selectively retains more predictable components. The gated outputs from all branches are aggregated to produce the final forecast.

Step 3: Results Analysis.

Quantile Regression is applied to generate forecasts at multiple quantile levels. The 50th percentile (median) represents the point forecast, equivalent to a conventional deterministic prediction [32]. Prediction intervals are computed at 50%, 70%, and 90% Prediction Interval Nominal Confidence (PINC) levels [33], allowing uncertainty characterization.

Model performance is evaluated from both deterministic and probabilistic perspectives. For point forecasting, the coefficient of determination (R^2), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE) are used to quantify goodness-of-fit and prediction accuracy. For interval forecasting, the Prediction Interval Normalized Average Width (PINAW) and the Average Interval Score (AIS) are used to assess the compactness and reliability of the prediction intervals. Together, these evaluation metrics provide a comprehensive assessment of the model's accuracy, uncertainty representation, and practical

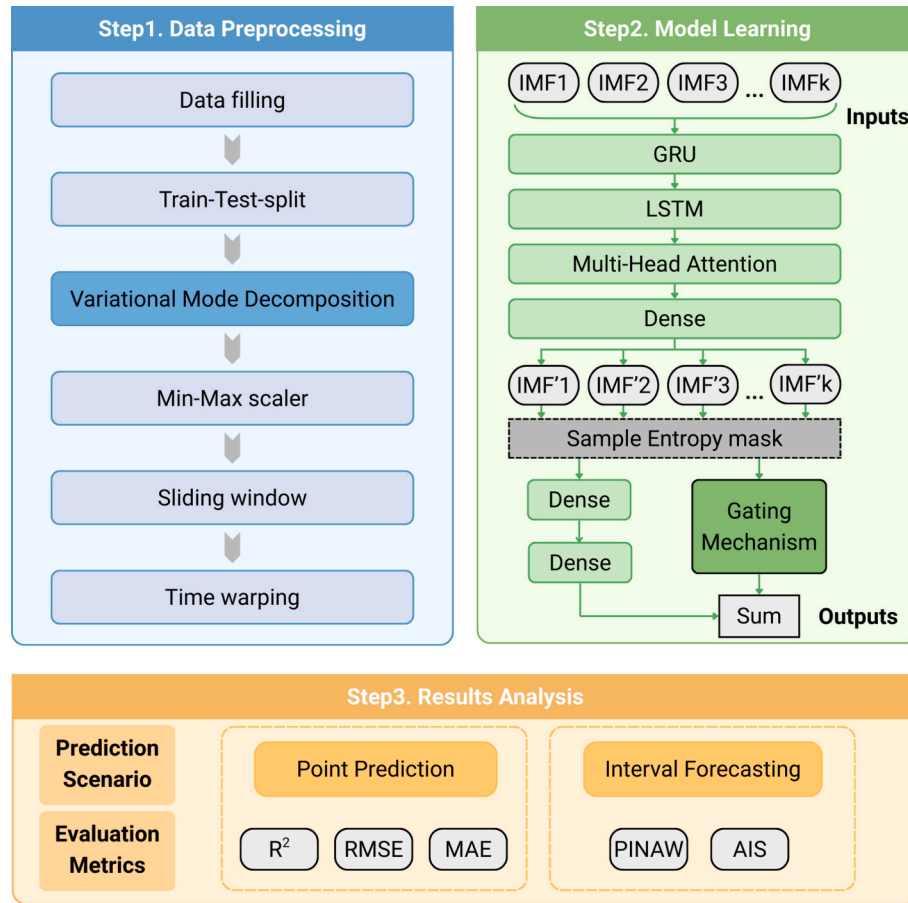


Fig. 1. Overall research framework of the proposed VMD-Attentive Quantile Network, illustrating data preprocessing, model learning, and probabilistic forecasting analysis.

applicability in real-world building electricity demand forecasting scenarios.

2.2. Case study Description

The empirical analysis in this study focuses on three campus buildings at National Taiwan University that represent distinct functional building types with comparable external conditions. Specifically, the selected buildings include a research-and-teaching building (Civil Engineering Research Building, CERB), an office-oriented academic building (Civil Engineering Department Building, CEB), and a mixed-use building integrating dining, teaching, and laboratory spaces (Ming-Da Hall, MDH). These buildings share the same climatic conditions, electricity pricing framework, and metering infrastructure, allowing controlled comparison across heterogeneous operational patterns while isolating differences attributable to building function.

The research-and-teaching building (CERB) exhibits pronounced intraday variability driven by extended working hours and research-related activities. The office-oriented building (CEB) shows relatively regular consumption patterns with clear weekday-weekend contrasts. The mixed-use building (MDH) combines a stable base load with intermittent short-term peaks during active periods. Together, these three functional types span a representative spectrum of building electricity demand behaviors, facilitating a systematic evaluation of model robustness and generalization under diverse temporal dynamics.

Hourly electricity consumption data are obtained from the university's energy monitoring and management system [34]. The dataset spans from January 1, 2019, 00:00 to December 31, 2023, 23:00 (UTC + 8), with electricity usage recorded in kilowatt-hours (kWh). This yields a

total of 43,824 hourly observations per building, providing a sufficiently long, high-resolution dataset for both point and probabilistic load forecasting.

2.3. Data Preparation

Data preprocessing is designed to ensure data quality, prevent information leakage, and preserve the intrinsic multi-scale characteristics of building electricity demand. The preprocessing pipeline begins with temporal alignment and data cleaning. Time zones and timestamps are first synchronized, after which invalid records caused by meter maintenance, power outages, or clearly abnormal readings are identified. In this study, abnormal readings are operationally defined as negative values and extreme spikes exceeding three standard deviations above the mean of each building's load series. Missing values are reconstructed using linear interpolation to preserve local temporal continuity. The longest consecutive missing segment does not exceed 12 h, and the proportion of removed records ranges from 1.76% to 2.30% across the building datasets. After quality control, the data are split chronologically into training, validation, and test sets at an 8:1:1 ratio. This temporal split ensures that all model evaluation reflects realistic forecasting conditions.

Each building's electricity demand series is then separately decomposed using VMD within each subset, rather than on the full time series, to avoid future information leakage and to extract multi-scale IMFs. This design ensures that no information from the validation or test sets is introduced into the training process during decomposition. This step reduces noise and isolates frequency-dependent patterns while preserving structural characteristics of the original signal. Detailed

mathematical formulations of VMD and the procedure for determining the number of IMFs are provided in the Supplementary Information (Sections S1–S2). Following decomposition, each IMF is normalized using Min–Max scaling to the range [1]. Importantly, normalization parameters are estimated exclusively from the training set and applied consistently to the validation and testing sets to avoid statistical leakage.

A sliding-window strategy is adopted [35], using the preceding 48 h of electricity consumption as input to predict demand over the subsequent 24 h. To further enhance robustness against temporal misalignment and local distortions, time warping–based data augmentation is applied to the training set [36]. For each training sample, each IMF channel is independently warped using a smooth temporal mapping generated by cubic spline interpolation, with random perturbations drawn from a normal distribution ($\mu = 1.0$, $\sigma = 0.02$) and $\text{knot} = 4$. The mapping is converted into a monotonic warping path through cumulative summation and normalization, and the warped sequence is resampled onto the original time grid using linear interpolation. The augmented samples are then concatenated with the original training data for model training.

Overall, the preprocessing workflow follows a sequential procedure comprising data cleaning, temporal splitting, signal decomposition, normalization, and sample generation with augmentation. All transformation parameters and hyperparameters are determined solely from the training data to mitigate the risk of data leakage, while maintaining the multi-scale structure and learnability of building electricity load series.

2.4. Model formulation

This section presents the formulation of the proposed VMD-Attentive Quantile Network, which is designed to perform probabilistic electricity demand forecasting by jointly modeling multi-scale temporal dynamics and predictive uncertainty. The model integrates a hierarchical GRU–LSTM temporal encoder, a Multi-Head Attention (MHA) module for temporal weighting, an IMF-dependent gating strategy for noise suppression, and a quantile regression framework trained with the Pinball loss.

2.4.1. Temporal encoding with GRU–LSTM

To capture temporal dependencies at multiple time scales, the proposed framework adopts a sequential GRU–LSTM architecture as its temporal encoder. The GRU is employed as the front-end module due to its relatively compact parameterization and fast convergence, making it effective for modeling short-term and local fluctuations in building electricity demand [37,38]. By regulating information flow through update and reset gates, the GRU adaptively balances historical memory retention and responsiveness to new inputs.

The GRU outputs are subsequently passed to an LSTM layer, which is designed to integrate longer-term dependencies through an explicit memory cell and gated information flow [39]. This hierarchical design enables the model to jointly represent short-term variability and long-term temporal trends commonly observed in building electricity consumption time series.

The detailed mathematical formulations and schematic structures of the GRU and LSTM units are provided in the Supplementary Information (Section S3) for completeness and reproducibility.

2.4.2. Multi-Head attention (MHA) for temporal importance weighting

Although recurrent neural networks can capture sequential dependencies, they do not explicitly differentiate the relative importance of individual time steps. To address this limitation, a Multi-Head Attention (MHA) mechanism derived from the Transformer architecture is applied along the temporal dimension [40].

The GRU–LSTM encoder outputs a sequence of hidden states, which are linearly projected into query (Q), key (K), and value (V) representations for each attention head. Scaled dot-product attention is then

computed as [41]:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

where d_k denotes the dimensionality of the key and query vectors in each attention head. The outputs from multiple attention heads are concatenated and linearly fused to form an attention-enhanced representation, which is then aggregated along the temporal axis to produce a global feature representation shared across all IMF branches.

By allowing different attention heads to focus on distinct temporal patterns—such as periodic behavior, abrupt changes, and delayed effects—the MHA module enhances the model’s ability to capture multi-scale temporal dependencies that are critical for accurate load forecasting.

The multi-head attention (MHA) module is positioned after the GRU–LSTM encoder to ensure that attention is applied to context-enriched temporal representations rather than raw input sequences [42]. The GRU and LSTM layers first encode the input series into hidden states that capture both short-term fluctuations and longer-term temporal dependencies, thereby forming context-aware sequential features. Applying attention on top of these hidden states enables the model to adaptively assign importance weights to different time steps based on their contributions to the forecasting task.

If attention were applied directly to the raw input sequence, each time step would contain only the original observations without sufficient contextual information, making it more difficult to distinguish meaningful temporal signals from local numerical variations. By placing the MHA module after the GRU–LSTM encoder, the attention mechanism operates on richer temporal representations, allowing the model to more effectively identify informative time steps and improve overall forecasting performance.

2.4.3. Imf-dependent gating strategy

After attention-based feature aggregation, the model employs IMF-dependent branches to improve robustness against noise and low-predictability components. In each branch, a learnable gating mechanism dynamically adjusts feature-wise contributions, suppressing noisy dimensions while retaining informative signals.

Specifically, the shared feature representation is distributed to the k -th IMF branch to form a branch-specific feature vector $u_t^{(k)}$. Based on this representation, a sigmoid-activated fully connected layer generates a gating vector $g_t^{(k)} \in [0, 1]^d$, while a parallel linear transformation produces a candidate feature representation $r_t^{(k)}$. The branch output is then obtained via elementwise multiplication between these two components:

$$g_t^{(k)} = \sigma\left(W_g^{(k)} u_t^{(k)} + b_g^{(k)}\right) \quad (2)$$

$$r_t^{(k)} = W_r^{(k)} u_t^{(k)} + b_r^{(k)} \quad (3)$$

$$o_t^{(k)} = g_t^{(k)} \odot r_t^{(k)} \quad (4)$$

where $W_g^{(k)}$ and $b_g^{(k)}$ denote the weight matrix and bias of the gating transformation, while $W_r^{(k)}$ and $b_r^{(k)}$ correspond to those of the candidate feature transformation in the k -th branch. The sigmoid function $\sigma(\cdot)$ maps gating values to $[0, 1]$, enabling adaptive feature selection. The operator $\odot$ denotes elementwise multiplication, through which $g_t^{(k)}$ modulates $r_t^{(k)}$ to produce the final branch output $o_t^{(k)}$.

To determine which IMF branches should activate gating, a sample entropy–based screening strategy is adopted to identify components with higher irregularity and lower predictability (see the details in Section S4 of the Supplementary Information). Based on these results, gating is activated only for the last two IMFs.

The mask-and-gating mechanism is placed in the later stage of the

network because its primary role is representation refinement rather than temporal dependency modeling. After the GRU-LSTM encoder and the attention module extract context-aware temporal representations, the gating mechanism selectively suppresses noisy IMF components according to their entropy levels, thereby improving the robustness of the learned representations.

2.4.4. Loss function for quantile forecasting

Quantile regression aims to characterize the conditional distribution of a target variable by directly estimating its quantiles, rather than producing a single point estimate. This formulation enables the model to capture asymmetric uncertainty and variability in electricity demand, which is particularly important for probabilistic load forecasting. Accordingly, the proposed framework adopts the Pinball Loss as the training objective to support both point forecasting and quantile-based interval prediction.

The Pinball Loss imposes asymmetric penalties on overestimation and underestimation errors, depending on the specified quantile level. By doing so, it encourages the model to learn accurate conditional quantile functions and simultaneously approximate multiple quantile curves. A lower Pinball Loss indicates not only improved point accuracy at a given quantile but also better calibration and reliability of the resulting prediction intervals, making it a widely used metric in probabilistic forecasting applications [43].

For a given quantile $q \in (0, 1)$, the Pinball Loss at time t is defined as [44]:

$$L_{q,t}(y_t, \hat{y}_t^q) = \begin{cases} (1-q)(\hat{y}_t^q - y_t) & \hat{y}_t^q \geq y_t \\ q(y_t - \hat{y}_t^q) & \hat{y}_t^q < y_t \end{cases} \quad (5)$$

where y_t denotes the observed load at time t , $\hat{y}_t^q$ is the predicted value of the q -th conditional quantile, and $L_{q,t}$ represents the corresponding loss contribution.

In this study, point forecasts are obtained from the median prediction by setting $q = 0.5$, while additional quantile outputs are used to construct prediction intervals at different confidence levels. This unified loss formulation allows the model to jointly optimize point accuracy and uncertainty representation within a single training framework.

2.5. Evaluation metrics

2.5.1. Metrics for point forecasting performance

To evaluate point forecasting performance, three widely used accuracy metrics are adopted: the coefficient of determination (R^2), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE). The coefficient of determination R^2 measures the proportion of variance in the observed data that is explained by the model; values closer to 1 indicate stronger explanatory power and better predictive performance [45]. MAE quantifies the average magnitude of absolute errors between predicted and observed values, providing a direct and interpretable measure of forecast accuracy [46]. RMSE computes the square root of the mean squared error and is more sensitive to large deviations, thereby emphasizing the impact of substantial prediction errors [47]. The corresponding formulations are given as [48]:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (6)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (7)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (8)$$

where n denotes the number of samples, y_i and $\hat{y}_i$ represent the observed and predicted values, respectively, and $\bar{y}$ is the mean of the

observed values.

2.5.2. Metrics for probabilistic and interval forecasting

For probabilistic forecasting, interval prediction quality is assessed using the Prediction Interval Normalized Average Width (PINAW) and the Average Interval Score (AIS). PINAW measures the average width of the prediction intervals normalized by the range of the observed data, reflecting the compactness of the forecast intervals under a given nominal coverage level [49]. Smaller PINAW values indicate tighter prediction intervals while maintaining appropriate coverage.

AIS provides a comprehensive evaluation by jointly considering interval width and coverage reliability. When the observed value lies within the prediction interval, the score reduces to the interval width; when it falls outside the interval, an additional penalty proportional to the distance from the interval bounds is imposed. As a result, AIS explicitly balances the trade-off between narrow intervals and adequate coverage, with lower AIS values indicating better overall interval forecasting performance. The two metrics are defined as [50]:

$$PINAW = \frac{1}{N} \sum_{i=1}^N \frac{U_i - L_i}{y_{\max} - y_{\min}} \quad (9)$$

$$AIS = \frac{1}{N} \sum_{i=1}^N \left[(U_i - L_i) + \frac{2}{\alpha} (\max(L_i - y_i, 0) + \max(y_i - U_i)) \right] \quad (10)$$

where N denotes the total number of samples; U_i and L_i are the upper and lower bounds of the prediction interval, respectively; y_i represents the observed value; and $y_{\max}$ and $y_{\min}$ denote the maximum and minimum observed values. The parameter α controls the penalty applied to observations falling outside the prediction interval. In this study, α is set to 0.1, 0.3, and 0.5 for the 90%, 70%, and 50% prediction intervals, respectively.

3. Results and discussion

3.1. Point prediction results

3.1.1. Cross-model comparison of point forecasting performance

This subsection evaluates the point forecasting performance of eight benchmark models that incorporate VMD as a preprocessing step: VMD-LSTM, VMD-GRU, VMD-RNN, VMD-CNN-LSTM, VMD-CNN-GRU, VMD-TA-Bi-LSTM, VMD-Transformer, and the proposed VMD-Attentive Quantile Network. The comparison is intentionally restricted to VMD-based models so that different forecasting architectures can be evaluated under a common decomposition-based signal representation, thereby isolating the effect of model design from preprocessing differences. To ensure a fair comparison, all benchmark models are constructed using a consistent preprocessing and experimental pipeline. Specifically, all models are trained on the same cleaned, decomposed, and normalized datasets, follow the same input-output sample generation scheme (i.e., identical sliding-window settings), and are evaluated under the same chronological train/validation/test partition and evaluation protocol. Their hyperparameters are determined based on validation performance using a consistent tuning principle across models. A sensitivity analysis of key hyperparameters is provided in the Supplementary Information (Section S5), demonstrating the robustness of the proposed model to moderate variations in parameter settings. For the proposed model, the median (50th percentile) output of the quantile regression framework is adopted as the point forecast. The evaluation metrics include R^2 , MAE, and RMSE. The corresponding hyperparameter settings of the proposed model are summarized in Table 1, and the complete comparative results are presented in Fig. 2.

Across all three buildings, the proposed model consistently achieves the highest R^2 and the lowest MAE and RMSE among the compared methods, indicating superior accuracy and robustness in point prediction.

Table 1

Hyperparameter settings of the proposed VMD-Attentive Quantile Network.

Category	Hyperparameter	Value
VMD configuration	Number of modes	7 (CERB, MDH); 8 (CEB)
	Penalty factor	2000
	Convergence tolerance	1×10^{-7}
	GRU hidden units	64
Neural network architecture	LSTM hidden units	128
	Attention embedding dimension	128
	Learning rate	0.001
Training settings	Batch size	32
	Number of training epochs	100

For the research-and-teaching building (CERB), the proposed model attains an R^2 of 0.808, an MAE of 5.170, and an RMSE of 7.291, corresponding to average improvements of 8.44%, 16.87%, and 13.48%, respectively, relative to other models. The relatively lower R^2 value for CERB highlights the inherent difficulty in forecasting research-oriented facilities, which are characterized by pronounced intraday variability and irregular consumption peaks driven by extended working hours and diverse research activities. These stochastic human behaviors introduce stronger non-stationary effects into the electricity demand signal, making precise tracking more challenging.

In the office-oriented academic building (CEB), the model demonstrates particularly robust performance, achieving an R^2 of 0.922, an MAE of 1.582, and an RMSE of 2.311, representing average improvements of 11.42%, 40.83%, and 39.57%. This exceptional accuracy stems from the highly regular and periodic consumption patterns typical of office-oriented usage, featuring clear weekday-weekend contrasts and synchronized operational schedules. Such structured signals are more effectively captured by the VMD-based multi-scale learning framework.

For the mixed-use building (MDH), which exhibits more complex and irregular operational patterns, the proposed model achieves an R^2 of 0.852, an MAE of 5.107, and an RMSE of 7.287, outperforming benchmark models by average margins of 12.85%, 24.45%, and 19.00%, respectively. The predictive performance for MDH reflects the model's capability to handle highly heterogeneous load patterns, as it must reconcile the stable base load from dining services with the intermittent, high-intensity peaks originating from laboratory and teaching spaces.

A closer examination of the comparative results reveals that VMD-RNN performs noticeably worse across all buildings, suggesting that conventional RNN architectures are insufficient for capturing both short-term fluctuations and longer-term dependencies in building electricity demand. Under the data scale and sliding-window configuration used in this study, VMD-Transformer and VMD-TA-Bi-LSTM exhibit

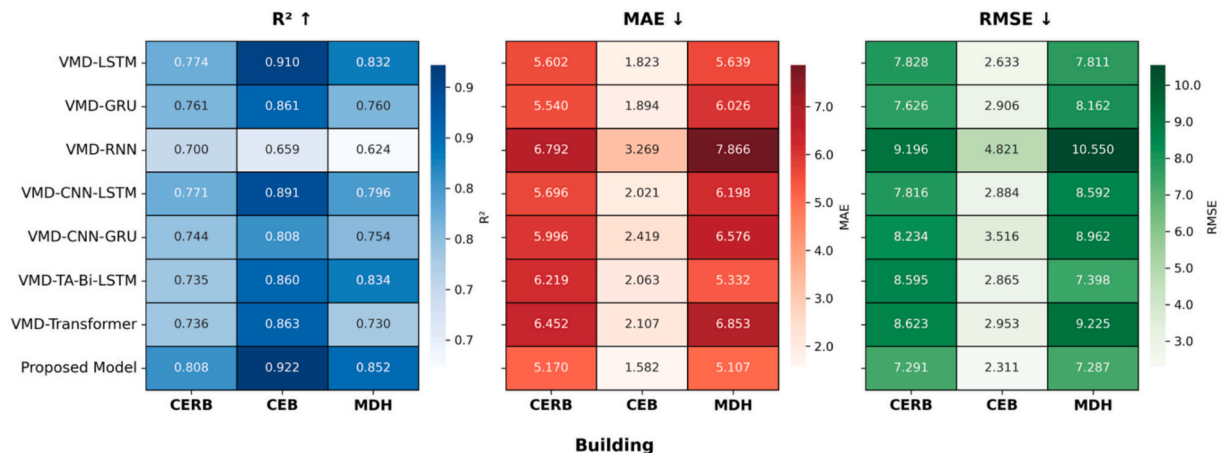
competitive performance in some cases, yet still yield higher error metrics than the proposed model. Overall, the results indicate that the integration of VMD-based multi-scale decomposition, learnable gating for noise suppression, and the complementary strengths of GRU and LSTM—combined with the use of the median quantile for point prediction—substantially enhances forecasting accuracy across heterogeneous building types. These findings confirm the effectiveness and general applicability of the proposed architecture for short-term building electricity demand forecasting.

3.1.2. Ablation study on point forecasting performance

To further investigate the contribution of individual submodules to point forecasting accuracy, an ablation study is conducted by systematically removing or modifying key components of the proposed architecture. The ablation variants include: (i) a baseline GRU–LSTM model without VMD, attention, or gating; (ii) two-layer recurrent variants, denoted as VMD-Gated-GRU and VMD-Gated-LSTM; (iii) a De-Gated model that removes the IMF-dependent gating mechanism; (iv) a De-MHA model that excludes the multi-head attention module; and (v) a VMD-Gated-LSTM-GRU model, in which the sequential order of LSTM and GRU is reversed. These variants are compared against the full VMD-Attentive Quantile Network. As in the cross-model comparison, the median (50th percentile) output is used as the point forecast, and the complete results are shown in Fig. 3.

The ablation results demonstrate that the baseline GRU–LSTM model without frequency decomposition, attention, or gating exhibits substantial performance degradation across all buildings. This degradation is particularly pronounced in the office-oriented academic building (CEB), where the baseline model achieves an R^2 of only 0.508 and an RMSE of 11.259, highlighting the critical role of VMD-based decomposition in mitigating nonstationary effects and improving data learnability. Although the two-layer recurrent variants (VMD-Gated-GRU and VMD-Gated-LSTM) benefit from deeper temporal modeling and gating, they still consistently underperform relative to the full model. For instance, in the mixed-use building (MDH), their R^2 values reach only 0.725 and 0.749, respectively, compared with 0.852 for the proposed architecture.

These results underline the complementary nature of the sequential GRU–LSTM design. The GRU efficiently captures short-term and localized variations, while the LSTM provides a more stable integration of longer-term dependencies. Reversing the order of the two components (VMD-Gated-LSTM-GRU) leads to inferior performance, indicating that the original configuration better aligns with the multi-scale temporal characteristics of building electricity demand. In addition, removing the IMF-dependent gating mechanism (De-Gated) results in increased prediction errors, confirming the effectiveness of learnable gating in

**Fig. 2.** Comparison of point forecasting performance across different models for the three buildings.

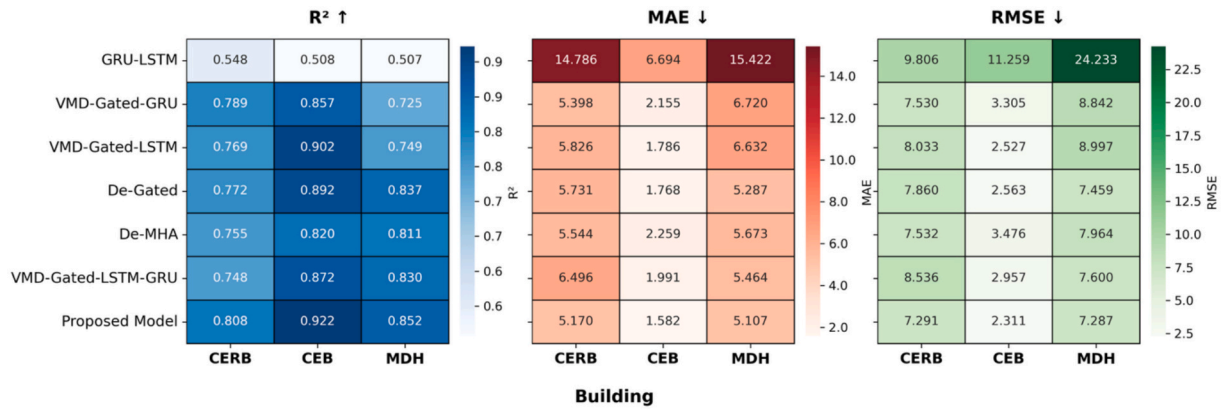


Fig. 3. Ablation study results for point forecasting performance of the proposed VMD-Attentive Quantile Network. *Note: Multi-head attention is not included in the GRU-LSTM and De-MHA variants.*

suppressing interference from high-entropy components. The removal of multi-head attention (De-MHA) leads to an even more pronounced decline in performance: averaged across the three buildings, R^2 decreases by 7.48%, while MAE and RMSE increase by 20.37% and 21.00%, respectively. This observation highlights the importance of temporal attention in emphasizing informative time periods and

reducing forecasting errors.

Overall, the ablation study provides clear evidence that each submodule—VMD decomposition, sequential GRU-LSTM encoding, multi-head attention, and IMF-dependent gating—contributes meaningfully to the accuracy and robustness of point forecasting. Their integration within a unified framework is therefore essential for

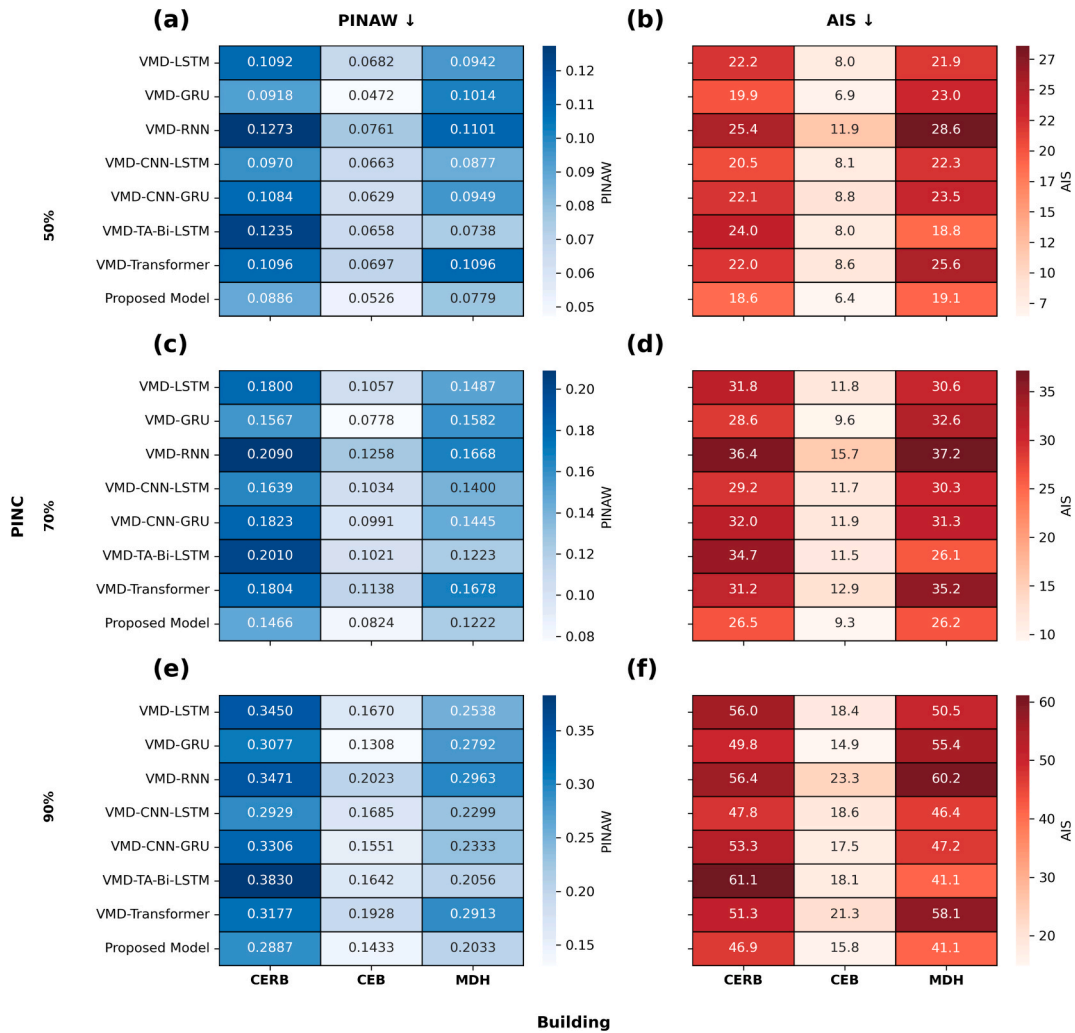


Fig. 4. Cross-model comparison of interval forecasting performance under different prediction interval nominal coverages (PINCs): (a) PINAW at 50% PINC, (b) AIS at 50% PINC, (c) PINAW at 70% PINC, (d) AIS at 70% PINC, (e) PINAW at 90% PINC, and (f) AIS at 90% PINC. Each subfigure compares the results of different forecasting models across CERB, CEB, and MDH.

achieving the superior performance observed in the proposed model.

3.2. Interval forecasting results

3.2.1. Cross-model comparison of interval forecasting performance

This subsection evaluates the interval forecasting performance of the same eight VMD-based models examined in Section 3.1.1, focusing on their ability to quantify predictive uncertainty under different prediction interval nominal coverages (PINCs), specifically 50%, 70%, and 90%. Performance is assessed using PINAW and AIS, which jointly reflect interval compactness and reliability.

As summarized in Fig. 4, the proposed VMD-Attentive Quantile Network consistently delivers strong and stable interval forecasting performance across the three campus buildings. For the research-and-teaching building (CERB), the proposed model achieves the lowest AIS values and competitive PINAW across all three PINCs, indicating a favorable balance between interval width and coverage reliability. For the office-oriented academic building (CEB), although VMD-GRU attains slightly narrower intervals (lower PINAW) at the 50% and 70% levels,

the proposed model yields lower AIS values, suggesting better-calibrated uncertainty estimates with fewer coverage violations. For the mixed-use building (MDH), while VMD-TA-Bi-LSTM marginally outperforms the proposed model at the 50% interval, the proposed model demonstrates superior stability and lower AIS at the 70% and 90% levels, which are more critical for risk-aware applications.

Overall, these results indicate that the proposed framework achieves a well-balanced trade-off between interval sharpness and coverage, particularly at higher confidence levels where reliable uncertainty quantification is most challenging. This robustness highlights the advantage of integrating VMD-based multi-scale decomposition, attention-driven temporal weighting, and quantile regression within a unified architecture.

To further illustrate the temporal behavior of probabilistic forecasts, Fig. 5 presents the hourly interval forecasting results of the proposed model for the three buildings. The median prediction (50th quantile) is shown by the blue curve, while the red curve denotes the observed electricity demand. Shaded bands represent the 50%, 70%, and 90% prediction intervals.

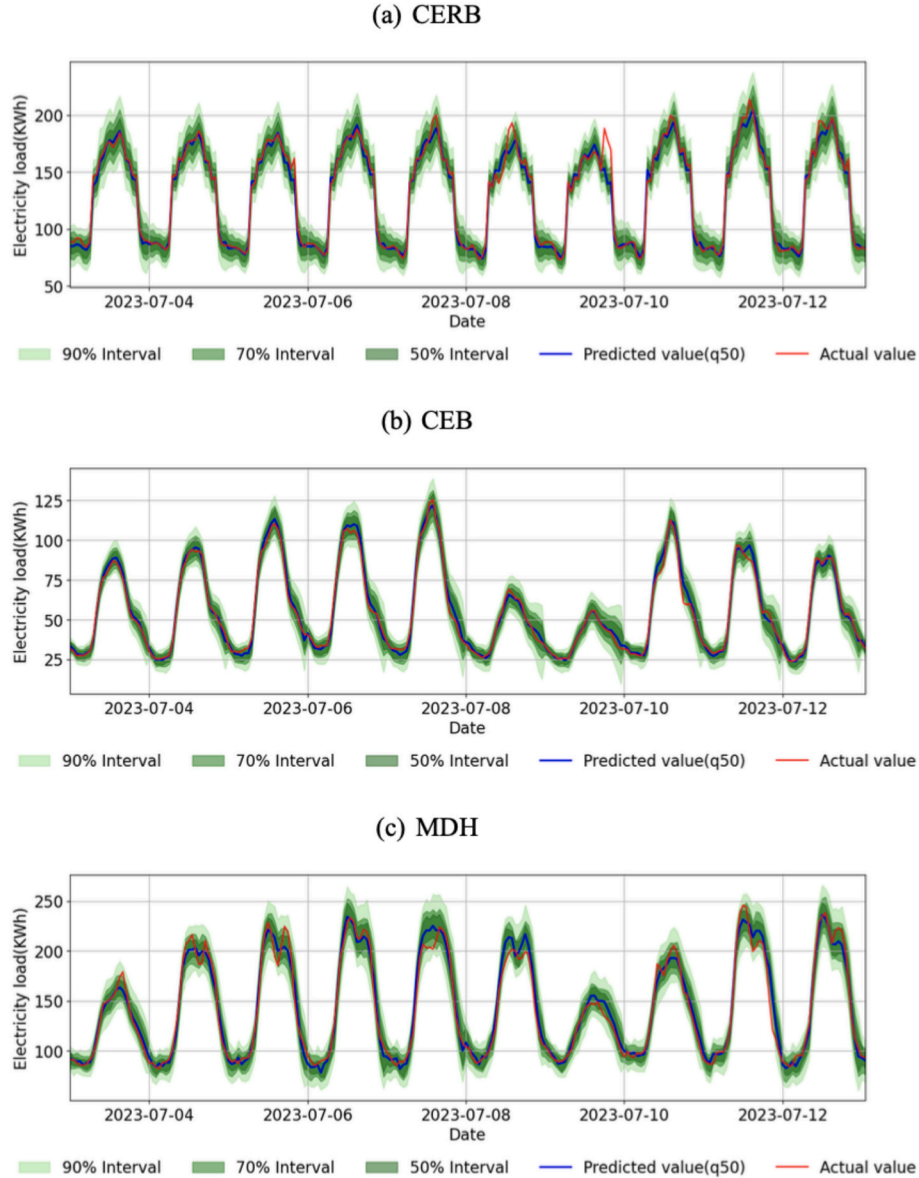


Fig. 5. Hourly probabilistic forecasting results of the proposed VMD-Attentive Quantile Network for the three campus buildings. Shaded regions indicate the 50%, 70%, and 90% prediction intervals.

As shown, the majority of observed values fall within the 90% prediction interval, indicating stable coverage across different operational conditions. The median prediction closely tracks the actual load, successfully capturing daily cycles, abrupt demand changes, and transient peak events. Importantly, the prediction intervals adapt dynamically over time: they widen during highly volatile periods and contract during stable phases, reflecting context-sensitive uncertainty estimation rather than static error bounds.

The three buildings exhibit distinct demand characteristics. CERB shows pronounced intraday variability with irregular peaks driven by mixed research and teaching activities. CEB presents more regular and periodic patterns typical of office-oriented usage. MDH, as a mixed-use facility, displays complex and rapidly changing load profiles. Across all cases, the proposed model maintains consistent interval coverage and adaptive width modulation, demonstrating its ability to generalize across heterogeneous consumption behaviors.

Together, these visualizations confirm that the VMD-Attentive Quantile Network not only performs well according to aggregate metrics but also produces interpretable, dynamically adaptive probabilistic forecasts suitable for practical building energy management applications.

3.2.2. Ablation study on interval forecasting performance

This subsection investigates how individual architectural components contribute to interval forecasting performance by conducting ablation experiments analogous to those in Section 3.1.2. The evaluated variants include the baseline GRU-LSTM, De-Gated, De-MHA, reversed-order GRU/LSTM models, and the full architecture. Results are summarized in Fig. 6 using PINAW and AIS across all PINCs.

For the research-and-teaching building (CERB), the GRU-LSTM baseline produces the narrowest intervals (lowest PINAW) but exhibits the highest AIS values, indicating poor coverage and unreliable uncertainty estimation. In contrast, the full model achieves lower AIS with moderately wider intervals, reflecting a more appropriate balance between sharpness and reliability. For the office-oriented academic building (CEB), simplified variants (De-Gated and De-MHA) achieve competitive performance at lower PINCs, suggesting that in highly regular demand patterns, reduced model complexity can still maintain adequate uncertainty calibration. However, these advantages diminish at the 90% interval. For the mixed-use building (MDH), which exhibits the most complex and volatile load behavior, the full model consistently achieves the lowest AIS and competitive PINAW across all PINCs, clearly outperforming all ablation variants.

Several key insights emerge from these results:

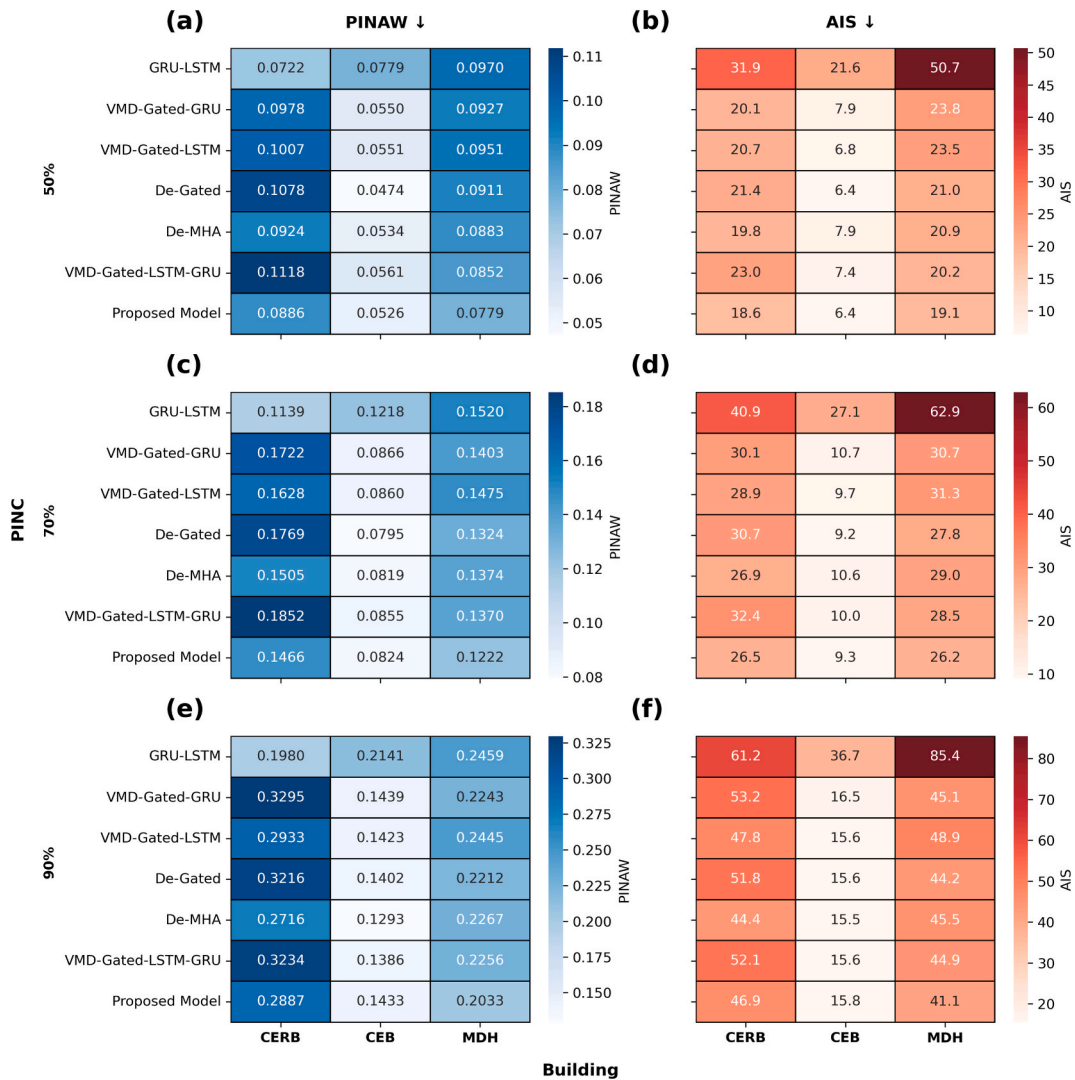


Fig. 6. Ablation study of interval forecasting performance for the proposed framework under different PINCs: (a) PINAW at 50% PINC, (b) AIS at 50% PINC, (c) PINAW at 70% PINC, (d) AIS at 70% PINC, (e) PINAW at 90% PINC, and (f) AIS at 90% PINC. Each subfigure compares the forecasting performance of different model variants across CERB, CEB, and MDH. *Note: Multi-head attention is not included in the GRU-LSTM and De-MHA variants.*

VMD-based multi-scale decomposition is critical for interval coverage, as removing it leads to systematically higher AIS values across buildings.

Narrower intervals alone do not guarantee better probabilistic forecasts—overly compact intervals often incur larger penalties due to insufficient coverage.

The benefits of gating and attention mechanisms are domain-dependent: while simplified variants may suffice for regular consumption patterns, the full architecture provides clear advantages in complex and high-variability environments.

Overall, the ablation study demonstrates that the proposed VMD-Attentive Quantile Network achieves more reliable and better-calibrated uncertainty estimates, particularly under challenging conditions where both sharpness and coverage must be balanced.

4. Conclusions

This study presents a probabilistic load forecasting framework based on the proposed VMD-Attentive Quantile Network, designed to jointly capture multi-scale temporal dynamics and predictive uncertainty in building electricity demand. By integrating Variational Mode Decomposition for frequency-band separation, a sequential GRU-LSTM encoder for complementary short- and long-term temporal modeling, Multi-Head Attention for dynamic temporal weighting, and an IMF-dependent gating mechanism for suppressing noise-dominated components, the framework provides a unified architecture for both point and interval forecasting. Quantile regression is incorporated to enable direct estimation of conditional demand distributions rather than relying on post hoc uncertainty assumptions.

The proposed framework is evaluated using hourly electricity consumption data from 2019 to 2023 for three functionally distinct buildings on the National Taiwan University campus, including a research-and-teaching building (CERB), an office-oriented academic building (CEB), and a mixed-use building (MDH). Across all three sites, the model consistently outperforms a diverse set of VMD-based benchmark models in point forecasting, as measured by R^2 , MAE, and RMSE. The observed improvements demonstrate that combining multi-scale decomposition with attention-enhanced temporal encoding and adaptive gating substantially improves forecasting accuracy and robustness under heterogeneous building usage patterns.

Beyond point prediction, the proposed VMD-Attentive Quantile Network demonstrates strong performance in probabilistic forecasting. Across multiple prediction confidence levels, the model achieves a balanced trade-off between interval compactness and coverage reliability, avoiding overly narrow prediction intervals that fail to capture demand variability. The prediction intervals adapt dynamically to changing load conditions, expanding during periods of high volatility and contracting during stable phases. This adaptive behavior provides more informative uncertainty estimates for building energy management applications.

The consistent performance observed across office-oriented, research-oriented, and mixed-use buildings highlights the adaptability of the proposed framework to diverse operational characteristics and load dynamics. These results suggest that the model is not limited to the specific campus context studied here, but can be readily extended to other office, commercial, or residential building systems, providing valuable decision support for demand response, capacity planning, and operational scheduling in smart energy systems.

Several limitations and opportunities for future research remain. The selection of VMD parameters and IMF gating criteria relies on empirical rules based on reconstruction error and sample entropy, and could be further improved through adaptive or end-to-end learning strategies. The current benchmark comparison is restricted to VMD-based models and does not yet include representative non-VMD probabilistic or distributional forecasting approaches. Extending the comparison to such

models will be important for broader benchmarking against state-of-the-art methods. In addition, no explicit non-crossing constraint is imposed on the quantile outputs in the current implementation. Although quantile crossings occur infrequently, they remain a known issue in multi-quantile forecasting. The corresponding quantile-crossing check is reported in Section S6 of the Supplementary Information. Methods such as non-crossing constrained deep networks [51] and the monotone composite quantile regression neural networks [52] provide promising directions for addressing this limitation. Future work will therefore focus on improving quantile consistency, expanding cross-domain and multi-scale datasets, incorporating external drivers, and refining adaptive decomposition strategies. In parallel, broader comparisons with representative probabilistic baselines, such as quantile-regression Transformer models [33] and empirical-copula-based approaches [53], will be conducted to further strengthen the generalizability and applicability of the proposed framework.

CRedit authorship contribution statement

Chun-Hao Huang: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Yu-Shin Hu:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis. **I-Yun Lisa Hsieh:** Writing – review & editing, Supervision, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.enbuild.2026.117696>.

Data availability

Data will be made available on request.

References

- [1] U. N. Environment, "Global Status Report for Buildings and Construction 2024/2025 | UNEP - UN Environment Programme." Accessed: Apr. 25, 2026. [Online]. Available: <https://www.unep.org/resources/report/global-status-report-buildings-and-construction-20242025>.
- [2] Y.-H. Chien, I.-Y.-L. Hsieh, Enhancing Urban Energy Resilience: Vehicle-to-Grid (V2G) strategies within Electric Scooter Battery Swapping Ecosystems, *Sustain. Cities Soc.* 118 (Jan. 2025) 106004, <https://doi.org/10.1016/j.scs.2024.106004>.
- [3] G. Li, et al., An improved transfer learning strategy for short-term cross-building energy prediction using data incremental, *Build. Simul.* 17 (1) (Jan. 2024) 165–183, <https://doi.org/10.1007/s12273-023-1053-x>.
- [4] D. Mariano-Hernández, L. Hernández-Callejo, A. Zorita-Lamadrid, O. Duque-Pérez, and F. Santos García, "A review of strategies for building energy management system: Model predictive control, demand side management, optimization, and fault detect & diagnosis," *J. Build. Eng.*, vol. 33, p. 101692, Jan. 2021, doi: 10.1016/j.jobte.2020.101692.
- [5] M. Ghafoori, M. Abdallah, S. Kim, Electricity peak shaving for commercial buildings using machine learning and vehicle to building (V2B) system, *Appl. Energy* 340 (Jun. 2023) 121052, <https://doi.org/10.1016/j.apenergy.2023.121052>.
- [6] Y. Ding, Q. Wang, Z. Wang, S. Han, N. Zhu, An occupancy-based model for building electricity consumption prediction: a case study of three campus buildings in Tianjin, *Energy Build.* 202 (Nov. 2019) 109412, <https://doi.org/10.1016/j.enbuild.2019.109412>.

- [7] M. Bourdeau, X. Qiang Zhai, E. Nefzaoui, X. Guo, and P. Chatellier, "Modeling and forecasting building energy consumption: A review of data-driven techniques," *Sustain. Cities Soc.*, vol. 48, p. 101533, Jul. 2019, doi: 10.1016/j.scs.2019.101533.
- [8] Y. Sun, F. Haghighat, B.C.M. Fung, A review of the-state-of-the-art in data-driven approaches for building energy prediction, *Energy Build.* 221 (Aug. 2020) 110022, <https://doi.org/10.1016/j.enbuild.2020.110022>.
- [9] T. Ahmad, H. Chen, Y. Guo, J. Wang, A comprehensive overview on the data driven and large scale based approaches for forecasting of building energy demand: a review, *Energy Build.* 165 (Apr. 2018) 301–320, <https://doi.org/10.1016/j.enbuild.2018.01.017>.
- [10] L. Xu, S. Wang, R. Tang, Probabilistic load forecasting for buildings considering weather forecasting uncertainty and uncertain peak load, *Appl. Energy* 237 (Mar. 2019) 180–195, <https://doi.org/10.1016/j.apenergy.2019.01.022>.
- [11] J. Runge, R. Zmeureanu, Forecasting Energy Use in buildings using Artificial Neural Networks: A Review, *Energies* 12 (17) (Jan. 2019) 3254, <https://doi.org/10.3390/en12173254>.
- [12] Y.T. Chae, R. Hoeshe, Y. Hwang, Y.M. Lee, Artificial neural network model for forecasting sub-hourly electricity usage in commercial buildings, *Energy Build.* 111 (Jan. 2016) 184–194, <https://doi.org/10.1016/j.enbuild.2015.11.045>.
- [13] A.-D. Pham, N.-T. Ngo, T. T. Ha Truong, N.-T. Huynh, and N.-S. Truong, "Predicting energy consumption in multiple buildings using machine learning for improving energy efficiency and sustainability," *J. Clean. Prod.*, vol. 260, p. 121082, Jul. 2020, doi: 10.1016/j.jclepro.2020.121082.
- [14] N. Somu, G. Raman M R, and K. Ramamritham, "A deep learning framework for building energy consumption forecast," *Renew. Sustain. Energy Rev.*, vol. 137, p. 110591, Mar. 2021, doi: 10.1016/j.rser.2020.110591.
- [15] G. Li, Y. Wang, C. Xu, J. Wang, X. Fang, C. Xiong, BO-STA-LSTM: Building energy prediction based on a Bayesian optimized spatial-temporal attention enhanced LSTM method, *Dev. Built Environ.* 18 (Apr. 2024) 100465, <https://doi.org/10.1016/j.dibe.2024.100465>.
- [16] C. Xu, Y. Sun, A. Du, D. Guo, Quantile regression based probabilistic forecasting of renewable energy generation and building electrical load: a state of the art review, *J. Build. Eng.* 79 (Nov. 2023) 107772, <https://doi.org/10.1016/j.jobe.2023.107772>.
- [17] J. Yu, W.-S. Chang, Y. Dong, Building Energy Prediction Models and Related Uncertainties: A Review, *Buildings* 12 (8) (Aug. 2022) 1284, <https://doi.org/10.3390/buildings12081284>.
- [18] A. Fouquier, S. Robert, F. Suard, L. Stéphan, A. Jay, State of the art in building modelling and energy performances prediction: a review, *Renew. Sustain. Energy Rev.* 23 (Jul. 2013) 272–288, <https://doi.org/10.1016/j.rser.2013.03.004>.
- [19] C. Liu, Z. Su, X. Zhang, A data-driven evidential regression model for building hourly energy consumption prediction with feature selection and parameters learning, *J. Build. Eng.* 80 (Dec. 2023) 107956, <https://doi.org/10.1016/j.jobe.2023.107956>.
- [20] E. Lucas Segarra, G. Ramos Ruiz, and C. Fernández Bandera, "Probabilistic Load Forecasting for Building Energy Models," *Sensors*, vol. 20, no. 22, p. 6525, Jan. 2020, doi: 10.3390/s20226525.
- [21] J. Roth, J. Chadlawada, R.K. Jain, C. Miller, Uncertainty matters: Bayesian Probabilistic forecasting for Residential Smart Meter Prediction, Segmentation, and Behavioral Measurement and Verification, *Energies* 14 (5) (Jan. 2021) 1481, <https://doi.org/10.3390/en14051481>.
- [22] C. S. Seefoo Jarquin, A. Gandelli, F. Grimaccia, and M. Mussetta, "Short-Term Probabilistic Load Forecasting in University Buildings by Means of Artificial Neural Networks," *Forecasting*, vol. 5, no. 2, pp. 390–404, Jun. 2023, doi: 10.3390/forecast5020021.
- [23] R. Liu, T. Chen, G. Sun, S.M. Muyeen, S. Lin, Y. Mi, Short-term probabilistic building load forecasting based on feature integrated artificial intelligent approach, *Electr. Power Syst. Res.* 206 (May 2022) 107802, <https://doi.org/10.1016/j.epsr.2022.107802>.
- [24] Y. Duan, A Novel Interval Energy-forecasting Method for Sustainable Building Management based on Deep Learning, *Sustainability* 14 (14) (Jan. 2022) 8584, <https://doi.org/10.3390/su14148584>.
- [25] N.E. Huang, et al., The empirical mode decomposition and the Hilbert spectrum for nonlinear and non-stationary time series analysis, *Proc. r. Soc. Math. Phys. Eng. Sci.* 454 (1971) (Mar. 1998) 903–995, <https://doi.org/10.1098/rspa.1998.0193>.
- [26] K. Dragomiretskiy, D. Zosso, Variational Mode Decomposition, *IEEE Trans. Signal Process.* 62 (3) (Feb. 2014) 531–544, <https://doi.org/10.1109/TSP.2013.2288675>.
- [27] I. Daubechies, The wavelet transform, time-frequency localization and signal analysis, *IEEE Trans. Inf. Theory* 36 (5) (Sep. 1990) 961–1005, <https://doi.org/10.1109/18.57199>.
- [28] L. Zhang, M. Alahmad, J. Wen, Comparison of time-frequency-analysis techniques applied in building energy data noise cancellation for building load forecasting: a real-building case study, *Energy Build.* 231 (Jan. 2021) 110592, <https://doi.org/10.1016/j.enbuild.2020.110592>.
- [29] S.-Y. Chou, A. Dewabharata, F.E. Zulfia, M. Fadil, Forecasting Building Energy Consumption using Ensemble Empirical Mode Decomposition, Wavelet Transformation, and Long Short-Term memory Algorithms, *Energies* 15 (3) (Jan. 2022) 1035, <https://doi.org/10.3390/en15031035>.
- [30] H. Zhang, M. Zhou, Y. Chen, W. Kong, Short-term power load forecasting for industrial buildings based on decomposition reconstruction and TCN-Informer-BiGRU, *Energy Build.* 347 (Nov. 2025) 116317, <https://doi.org/10.1016/j.enbuild.2025.116317>.
- [31] W. Chen, F. Rong, C. Lin, Short-term building electricity load forecasting with a hybrid deep learning method, *Energy Build.* 330 (Mar. 2025) 115342, <https://doi.org/10.1016/j.enbuild.2025.115342>.
- [32] Y. Hu, X. Deng, L. Yang, Interval prediction model for residential daily carbon dioxide emissions based on extended long short-term memory integrating quantile regression and sparse attention, *Energy Build.* 333 (Apr. 2025) 115481, <https://doi.org/10.1016/j.enbuild.2025.115481>.
- [33] W. Zhang, H. Zhan, H. Sun, M. Yang, Probabilistic load forecasting for integrated energy systems based on quantile regression patch time series Transformer, *Energy Rep.* 13 (Jun. 2025) 303–317, <https://doi.org/10.1016/j.egy.2024.11.057>.
- [34] "National Taiwan University. (2023). Campus Digital Electricity Meter Monitoring System (校園數位電錶監視系統)." [Online]. Available: https://epower.ga.ntu.edu.tw/?fbclid=IwAR1_crXmTrEojnqGZCh6z2hesnkZ1Bsdy7YBEnyAyzEYHOolvr-xjA8sBAqo.
- [35] D. N. Papadopoulos, F. Dadras Javan, B. Najafi, A. Haghighat Mamaghani, and F. Rinaldi, "Handling complete short-term data logging failure in smart buildings: Machine learning based forecasting pipelines with sliding-window training scheme," *Energy Build.*, vol. 301, p. 113694, Dec. 2023, doi: 10.1016/j.enbuild.2023.113694.
- [36] V. Mangé, J.-Y. Tournet, F. Vincent, L. Mirambell, and F. Manzoni Vieira, "Anomaly detection in ship trajectories using machine learning and dynamic time warping," *Eng. Appl. Artif. Intell.*, vol. 157, p. 111185, Oct. 2025, doi: 10.1016/j.engappai.2025.111185.
- [37] R. Quan, G. Cheng, X. Guan, G. Zhang, J. Quan, A HO-BiGRU-Transformer based PEMFC degradation prediction method under different current conditions, *Renew. Energy* 256 (Jan. 2026) 124132, <https://doi.org/10.1016/j.renene.2025.124132>.
- [38] A. M. Sathe, S. R., and A. A. Thomas, "Enhancing agricultural sustainability: Time series forecasting with ICEEMDAN-VMD-GRU for economic-resilience," *Results Eng.*, vol. 27, p. 106423, Sep. 2025, doi: 10.1016/j.rineng.2025.106423.
- [39] W. Liu, Y. Bai, X. Yue, R. Wang, Q. Song, A wind speed forecasting model based on rime optimization based VMD and multi-headed self-attention-LSTM, *Energy* 294 (May 2024) 130726, <https://doi.org/10.1016/j.energy.2024.130726>.
- [40] B. Lim, S.O. Arık, N. Loeff, T. Pfister, Temporal Fusion Transformers for interpretable multi-horizon time series forecasting, *Int. J. Forecast.* 37 (4) (Oct. 2021) 1748–1764, <https://doi.org/10.1016/j.ijforecast.2021.03.012>.
- [41] A. Vaswani et al., "Attention Is All You Need," Aug. 02, 2023, arXiv: arXiv:1706.03762, doi: 10.48550/arXiv.1706.03762.
- [42] G. Liu, J. Guo, Bidirectional LSTM with attention mechanism and convolutional layer for text classification, *Neurocomputing* 337 (Apr. 2019) 325–338, <https://doi.org/10.1016/j.neucom.2019.01.078>.
- [43] Y. Wang, D. Gan, M. Sun, N. Zhang, Z. Lu, C. Kang, Probabilistic individual load forecasting using pinball loss guided LSTM, *Appl. Energy* 235 (Feb. 2019) 10–20, <https://doi.org/10.1016/j.apenergy.2018.10.078>.
- [44] R. Koenker, G. Bassett, Regression Quantiles, *Econometrica* 46 (1) (1978) 33–50, <https://doi.org/10.2307/1913643>.
- [45] Y. Yang, Q. Fu, J. Chen, Y. Lu, Y. Wang, H. Wu, Temporal feature decomposition fusion network for building energy multi-step prediction, *J. Build. Eng.* 94 (Oct. 2024) 110034, <https://doi.org/10.1016/j.jobe.2024.110034>.
- [46] F. Mo, et al., A novel multi-step ahead solar power prediction scheme by deep learning on transformer structure, *Renew. Energy* 230 (Sep. 2024) 120780, <https://doi.org/10.1016/j.renene.2024.120780>.
- [47] C. Kanthila, A. Boodi, A. Marszał-Pomianowska, K. Beddier, Y. Amirat, M. Benbouzid, Enhanced multi-horizon occupancy prediction in smart buildings using cascaded Bi-LSTM models with integrated features, *Energy Build.* 318 (Sep. 2024) 114442, <https://doi.org/10.1016/j.enbuild.2024.114442>.
- [48] A. Ghasempour, "Support Vector Regression to Predict Power Consumption," The University of New Mexico, 2024. [Online]. Available: https://digitalrepository.unm.edu/ece_rpts/59.
- [49] X. Serrano-Guerrero, M. Briceño-León, J.-M. Clairand, G. Escrivá-Escrivá, A new interval prediction methodology for short-term electric load forecasting based on pattern recognition, *Appl. Energy* 297 (Sep. 2021) 117173, <https://doi.org/10.1016/j.apenergy.2021.117173>.
- [50] T. Gneiting, A.E. Raftery, Strictly proper Scoring Rules, Prediction, and Estimation, *J. Am. Stat. Assoc.* 102 (477) (Mar. 2007) 359–378, <https://doi.org/10.1198/016214506000001437>.
- [51] S. Lu, Q. Xu, C. Jiang, Y. Liu, A. Kusiak, Probabilistic load forecasting with a non-crossing sparse-group Lasso-quantile regression deep neural network, *Energy* 242 (Mar. 2022) 122955, <https://doi.org/10.1016/j.energy.2021.122955>.
- [52] A.J. Cannon, Non-crossing nonlinear regression quantiles by monotone composite quantile regression neural network, with application to rainfall extremes, *Stoch. Environ. Res. Risk Assess.* 32 (11) (Nov. 2018) 3207–3225, <https://doi.org/10.1007/s00477-018-1573-6>.
- [53] P.F. Austnes, C. García-Pareja, F. Nobile, M. Paolone, Probabilistic load forecasting of distribution power systems based on empirical copulas, *Sustain. Energy Grids Netw.* 42 (Jun. 2025) 101708, <https://doi.org/10.1016/j.segan.2025.101708>.